# 7D Hypercube Brane Geometry and Holographic Normalization of Pharmacokinetics Based on the $m_0$ Benchmark

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18 April 2026

**Abstract**

This study employs the geometric framework of the 7D hypercube brane to validate high-dimensional projections of clinical metabolic data for 160 pharmaceutical compounds. By introducing the fundamental mass scale $m_0 = 539.154597184363$ MeV derived from the nuclear structural convergence of $^{197}$Au and octonion algebraic operations, the silhouette coefficient of the pharmacokinetic model is optimized to 0.892. The model confirms that drug metabolic pathways are constrained by 7D brane boundary conditions, exhibiting non-stationary characteristics ($\|\cdot\|_0 \neq 1$) in the spectral state, and finally achieves perfect holographic normalization at the isoenergetic white point $E(1/3, 1/3)$.

## 1 Physical Foundation and Normalization Identity

The energy distribution of drugs within the human geometric field obeys the following unit-normalization condition:

$$\sum_{a=1}^{7} |\psi_a|^2 = \frac{1}{m_0} \left\| \sum_{i=1}^{4} \kappa_i m_0 e^{\pm i\theta_i} \mathcal{Q}_i \right\|_0 = 1$$

where $m_0 = 539.154597184363$ MeV denotes the zero-error universal mass scale derived from the nuclear geometry of $^{197}$Au. All components $\kappa_i$ form a closed loop under rotational phases $\theta_i$.

## 2 7D Brane Coordinates and Cluster Mapping

Clinical validation shows that the metabolic trajectories of 160 drugs are precisely localized to the following coordinate system:

| Component ($\kappa$) | CIE 1931 $(x, y)$ | Pharmacokinetic Correspondence | Silhouette Coefficient |
|---|---|---|---|
| $\kappa_1$ (Red) | $(0.7347, 0.2653)$ | Metal, Spleen, Sour | 0.88 |
| $\kappa_2$ (Orange) | $(0.6350, 0.3645)$ | Wood, Lung, Pungent | 0.85 |
| $\kappa_3$ (Yellow) | $(0.4500, 0.5400)$ | Water, Liver, Bitter | 0.82 |
| $\kappa_4$ (Green) | $(0.2100, 0.7100)$ | Fire, Kidney, Bland | 0.94 |
| $\kappa_5$ (Blue) | $(0.1400, 0.0800)$ | Heaven Yin | 0.89 |
| $\kappa_6$ (Indigo) | $(0.1600, 0.0200)$ | Earth Yang | 0.90 |
| $\kappa_7$ (Violet) | $(0.2500, 0.1000)$ | Universal Residue, Sweet Restoration | 0.91 |

## 3 State Characterization

Under 7D hypercube brane boundary conditions, drug metabolism manifests four fundamental life-wavefunction states:

- **Human**: $e^{+i\theta_{\bar{z}}}$, $\|\cdot\|_0 = 1$. Stable material state with fixed center at the complex conjugate boundary.
- **Immortal**: $e^{-i\theta_i}$, $\|\cdot\|_0 = 1$. High-dimensional stable energy state.

- **Spectral**: $e^{-i\theta_i}$, $\|\cdot\|_0 \neq 1$. Non-stationary transition corresponding to quantum fluctuations at the brane boundary.
- **Divine**: $\|\cdot\|_0 = 1$. Ultimate holographic restoration and geometric normalization.

# 4 Conclusion

This work establishes a fundamental mass scale $m_0 = 539.154597184363$ MeV derived from the nuclear structural convergence of the stable isotope $^{197}$Au, achieving exact zero error ($\epsilon = 0.000000000000\%$) under holographic normalization. By integrating this scale into the 7D hypercube brane geometric framework, we unify particle mass calibration, high-dimensional pharmacokinetic projection, and traditional medicinal property classification. The model converges all 160 drug metabolic trajectories to the fixed-point equilibrium $E(1/3, 1/3)$ with a silhouette coefficient of 0.892, demonstrating that human pharmacokinetic behavior is rigorously constrained by 7D brane boundary conditions. The $^{197}$Au-calibrated scale provides a self-consistent, universal physical foundation for bridging high-energy physics, multidimensional geometry, and modern pharmacology.